

Introduction to Inborn Errors of Immunity

William K. Dolen, MD • 54-slide current deck • phenotype-first diagnostic approach

5 official objectives

2 ScholarRx bricks

First / second / third line workup

IEI phenotypes

SCID T/B/NK table

SOURCE PRIORITY: the current 54-slide Dolen deck controls terminology, organization, testing strategy, and faculty emphasis. The Word guide is secondary. It is useful for classic disease details, but it simplifies the current deck and does not include the full phenotype framework, the “stealth objective,” or the full SCID immunophenotype table.

Official + Faculty Objectives

12.1 Recognize clinical warning signs and plan an evaluation when IEI is suspected.

12.2 Know first-line screening tests and second-line laboratory evaluation.

12.3 Understand genotyping for definitive diagnosis and carrier detection.

12.4 Recognize pathogenesis, clinical, and laboratory features of the major IEIs.

12.5 Classify major SCID forms by T-, B-, and NK-cell flow-cytometry phenotype and connect phenotype to genetic defect.

Faculty “stealth objective”: also recognize less-traveled IELs: ataxia-telangiectasia, MSMD, hyper-IgE syndromes, chronic mucocutaneous candidiasis / APS-1, IRAK4 or MyD88 deficiency, and UNC93B or TLR3 deficiency.

ASSIGNED SCHOLARRX

Introduction to Inborn Errors of Immunity, Part 1

Introduction to Inborn Errors of Immunity, Part 2

Slides 6-9: When Should You Suspect an IEI?

Clinical Warning Signs



Children

- Family history of *bona fide* IEI
- Abnormal newborn screen (TREC)
- ≥ 4 new ear infections in a year
- ≥ 2 serious sinus infections in a year
- ≥ 2 pneumonias in 1 year
- Recurrent deep skin or organ abscesses
- Persistent thrush, or fungal skin infection
- ≥ 2 episodes of sepsis or deep-seated infections
- ≥ 2 months on antibiotics with poor response
- Failure to thrive in an infant
- Need for IV antibiotics to treat infections

Adapted from: Jeffery Modell Foundation handout



Slide 7. Childhood warning signs. The lecture combines family history/newborn screening with recurrent or unusually severe infections, poor antibiotic response, persistent fungal disease, failure to thrive, and need for IV therapy.

Child warning sign	Why it matters in the deck's framework
Family history of bona fide IEI	Raises pretest probability immediately
Abnormal newborn TREC screen	Congenital T-cell lymphopenia must be evaluated
≥4 ear infections / year	Recurrent infection burden
≥2 serious sinus infections / year	Possible antibody / host-defense defect
≥2 pneumonias / year	Repeated serious lower-respiratory disease
Deep abscesses / deep-seated infection / sepsis	Think phagocyte or other major host-defense defect
Persistent thrush / fungal skin infection	Raises T-cell / fungal-host-defense concern
≥2 months antibiotics with poor response	Unusually difficult-to-clear infection
Failure to thrive	Must-not-miss infant clue
Need for IV antibiotics	Severity out of proportion to routine childhood infection

Evaluation Plan



- Detailed **history** of the **types of infections**
- Complete **physical exam** with focus on lymphatic system, and areas that have been affected by recurrent infections
- Think in terms of disease **phenotypes**
- A **screening lab evaluation** can (and should) be ordered in a primary care setting
- Narrow it down from there; consider using AI for help; **consult immunology for help**
- **Genotyping** available for nearly all diseases; usually ordered by immunology in consultation with pathology and genetic counseling
- Most patients will be 'normal', but new IEs await discovery



Slide 9. The evaluation plan starts with the infection pattern and complete examination, then asks you to identify a phenotype, order screening labs, narrow the differential, and involve immunology/genetics appropriately.

FACULTY APPROACH

History of infections → phenotype → screening labs → targeted second-line tests → genetics / specialist input.

Do not diagnose IEI by infection count alone. Slide 46 later includes a “normal child phenotype”: day care exposure, viral/bacterial diagnostic difficulty, allergic rhinitis mislabeled as sinusitis, and asthma mislabeled as bronchitis/pneumonia.

Slides 10-18: First-Line Screening - What You Should Be Able to Order

Screening Labs



Basic – first line – YOU can order and interpret these!

- CBC with differential
- Immunoglobulin levels (GAME)
- Maybe - delayed hypersensitivity skin testing (DHST)
- HIV-1 testing
- C3, C4, CH50, AH50 (may be mailouts)
- Check TREC screen results (was there neonatal lymphopenia?)



Slide 11. The faculty's first-line list: CBC with differential, immunoglobulin levels (GAME), possibly delayed-hypersensitivity skin testing, HIV-1 testing, C3/C4/CH50/AH50, and review of the newborn TREC result.

1. CBC with differential

CBC with Differential



- Always calculate **absolute numbers** of lymphocytes and neutrophils
- Look at platelet number **and size**; examine blood smear manually with a pathologist
- Could it be:
 - **Reticular dysgenesis**
 - Various types of neutropenia
 - **Chédiak-Higashi syndrome**
 - **SCID** (low absolute lymphocyte count)
 - **Wiskott-Aldrich syndrome: microthrombocytopenia**



Slide 12. CBC interpretation is not just 'normal/abnormal': calculate absolute lymphocyte and neutrophil counts and examine platelet number and size.

- **Low absolute lymphocyte count:** think SCID / T-cell lymphopenia.
- **Neutropenia:** consider neutrophil-number disorders.
- **Microthrombocytopenia:** points toward Wiskott-Aldrich syndrome.
- Manual smear review can reveal clues such as Chédiak-Higashi morphology.

2. Quantitative immunoglobulins - GAME

Immunoglobulin Levels



- Low IgG, IgA, IgM, IgE
 - All very low (near zero)? Think **SCID**
 - **X-linked agammaglobulinemia** 1 or 2
 - **Common variable immunodeficiency** phenotype (2 or more low classes with no alternative explanation)
- Normal or elevated IgM, with very low IgG, IgA, IgE
 - **Hyper IgM phenotype**
- Highly elevated IgE: is this atopic dermatitis or could it be **hyper IgE syndrome**?
- Very elevated Ig levels (hypergammaglobulinemia)
 - **HIV-1** disease
 - **Chronic granulomatous disease** (CGD)



Slide 13. The deck uses the quantitative Ig pattern itself to suggest phenotype: pan-low, isolated/multiclass low, hyper-IgM, high IgE, or hypergammaglobulinemia.

Ig pattern	Deck association
All Ig classes near zero	Think SCID
Very low immunoglobulins	XLA1/2 can fit
≥2 low classes without alternative explanation	“CVID” phenotype
Normal/high IgM + very low IgG, IgA, IgE	Hyper-IgM phenotype
Highly elevated IgE	Atopic dermatitis vs hyper-IgE syndrome
Hypergammaglobulinemia	HIV-1 disease or CGD can do this

3. TREC newborn screening

TRECs



- Part of routine newborn screen in all states
- V(D)J recombination causes DNA excision, forming stable extrachromosomal circles
- TRECs counted by real-time PCR, using cord blood or peripheral blood, including blood from Guthrie cards
- Detects congenital lymphopenia (SCID being one cause)
- For older infants and children, check the neonatal result
- TRECs not normally reordered unless the newborn screen was somehow equivocal



Slide 16. TRECs arise from DNA excision during V(D)J recombination and are counted by PCR. The screen detects congenital lymphopenia; SCID is one cause, not the only cause.

TREC TRAP

An abnormal TREC screen does **not** equal SCID. The deck lists SCID/leaky SCID, DiGeorge, ataxia-telangiectasia, other congenital syndromes, and secondary causes such as cardiac disease, GI loss, leukemia, and third spacing.

4. Complement screening

Complement



- **CH50**: screen of classic pathway
- **AH50**: screen of alternate pathway
- Also order **C3**, **C4** to help interpret results
 - Consumption?
 - Deficiency?
 - Lab error (prone to specimen mishandling)?
- Mannan-binding lectin
- C1 inhibitor (quantitative, functional)
- C1q: acquired angioedema



Slide 18. CH50 screens the classical pathway; AH50 screens the alternative pathway. C3 and C4 help distinguish deficiency, consumption, and possible specimen/lab issues.

COMPLEMENT FIRST-PASS

CH50 = classical • AH50 = alternative • add C3/C4.

For angioedema evaluation, the deck also names C1 inhibitor quantity/function and C1q for acquired angioedema.

5. HIV-1 testing / older DHST

- The lecture explicitly says HIV testing can be indicated even without traditional “risk factors.”
- Delayed-hypersensitivity skin testing is a traditional in-vivo T-cell-function screen that is fading from routine use.

Slides 19-24: Second-Line Tests

More Advanced – Second Line



Usually ordered by immunology

- Antibody functional studies
- Lymphocyte flow cytometry
- Mitogen and antigen stimulation
- Basic neutrophil studies



Slide 19. Second-line testing is usually ordered by immunology: antibody functional studies, lymphocyte flow cytometry, mitogen/antigen stimulation, and neutrophil studies.

Antibody function

- IgM: isohemagglutinin titers.
- IgG: titers to childhood immunizations or common infections.
- IgE: allergy testing.

Flow cytometry

Flow Cytometry



- Not part of a “screening evaluation”
- Many common and esoteric mAbs available for large number of cell surface markers
 - Beware of using esoteric markers
- Counting T cell subsets, B cells, and NK cells is clinically useful
 - SCID immunophenotyping
- **CD45RA/CD45RO**: naïve/memory T cells (somewhat esoteric)
- Also used for things like:
 - **CD40L** detection (X-linked hyper IgM)
 - Certain neutrophil function studies (**dihydrorhodamine** assay for neutrophil oxidative burst)



Slide 21. Flow is not part of the basic screening evaluation. Clinically useful outputs include T-, B-, and NK-cell counts for SCID phenotyping; the deck also names CD40L detection and DHR-related neutrophil work.

Screening vs confirmation: do not jump straight to giant esoteric flow panels. The deck explicitly warns against overusing esoteric markers.

Mitogen / antigen stimulation

Lymphocytes are cultured in vitro, exposed to nonspecific mitogens or specific antigens, and assessed for proliferation.

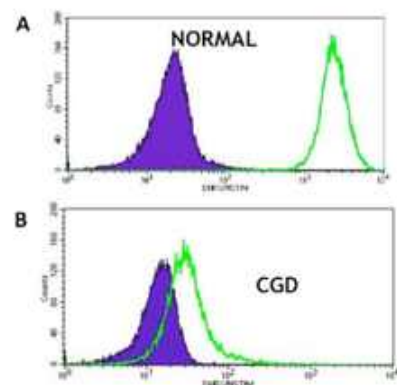
DHR / NBT for neutrophil oxidative burst

Neutrophil Studies



DHR (dihydrorhodamine) assay

- The blue flow cytometry histogram shows neutrophils at rest, and the green histogram shows the effect of *in vitro* stimulation of the respiratory burst
- The CGD patient has essentially no response
- A female carrier of X-linked CGD will show **two populations** following stimulation



Slide 23. DHR flow-cytometry example: normal neutrophils shift after stimulation; CGD shows essentially no oxidative-burst response. An X-linked female carrier can show two populations.

CGD TEST

DHR is the modern high-yield oxidative-burst test. NBT is older but still appears in books/question banks.

Slides 25-28: Third-Line / Genotyping

Genotyping



- Essential for definitive (vs. phenotypic) diagnosis
 - (e.g., 'CVID' is a phenotype, not a diagnosis)
- Essential for effective genetic counseling
 - Some IEIs have both autosomal and x-linked forms (e.g., CGD, hyper IgM syndrome)
 - Some AR IEIs present with clinical symptoms in the heterozygous state (e.g., DOCK8 form of hyper IgE syndrome)



Slide 27. The deck emphasizes that molecular testing changes a phenotypic label into a definitive diagnosis and is essential for genetic counseling.

"CVID" FACULTY NUANCE

"CVID" is a phenotype / working preliminary diagnosis, not necessarily the final molecular diagnosis.

- Strong suspicion: individual-gene test or targeted panel.
- Less clear / broad phenotype: whole-exome or whole-genome sequencing may be appropriate.
- Genetic counseling matters because disorders such as CGD and hyper-IgM have both X-linked and autosomal forms.

Companion simplification: the Word guide frames genotyping as the definitive diagnostic test for most IEIs. The current deck is more nuanced: use phenotype and

targeted testing first, then individual genes/panels/exome/genome depending on suspicion and specialist input.

Slides 29-33 and 37-46: Think in Phenotypes, Not a 500-Disease List

IEI Phenotypes



- Combined immunodeficiency (SCID) phenotype
- T cell dysfunction phenotype
- Hyper IgM phenotype
- Antibody deficiency phenotype
- Complement deficiency phenotype
- Phagocyte deficiency phenotype
- Normal child with too many infections



Slide 32. The lecture's organizing framework: combined/SCID, T-cell dysfunction, hyper-IgM, antibody deficiency, complement deficiency, phagocyte deficiency, and the normal child with too many infections.

FACULTY EXPLICITLY SAYS

Don't memorize >500 IEIs. Identify the infection phenotype, connect it to a host-defense mechanism, and then test that mechanism.

Combined IEI Phenotype



- All types of infections (viral, bacterial, fungal, protozoal) - the “walking agar plate”
- **Adult presentations reported**
- Low age-related total lymphocyte count
- **SCID** immunophenotypes
- Also: **ataxia-telangiectasia**



Slide 33. Combined IEI phenotype: broad susceptibility across viral, bacterial, fungal, and protozoal infections - the slide's memorable phrase is the 'walking agar plate.'

Combined / SCID

All pathogen classes; low age-adjusted lymphocyte count; SCID immunophenotype; ataxia-telangiectasia can overlap.

T-cell dysfunction

Recurrent viral, fungal, protozoal infections; DiGeorge is the clinical example; WAS, CMC/APS-1 and selected TLR-pathway defects appear in the list.

Hyper-IgM

Antibody-deficiency-like infections plus some fungi; normal/high IgM with other classes near zero; CD40L deficiency is faculty high-yield.

Antibody deficiency

Mostly bacterial infections; XLA, selective IgA deficiency, “CVID,” transient hypogammaglobulinemia of infancy.

Complement deficiency

Terminal pathway → recurrent *Neisseria*; other patterns include bacterial disease, lupus-like immune-complex disease, PNH, angioedema.

Phagocyte deficiency

Deep boils/abscesses, pneumonia, osteomyelitis; *Aspergillus*, *Nocardia*, *Serratia*, *Staphylococcus* suggest neutrophil dysfunction.

T Cell Dysfunction Phenotype



- Does your patient have problems with recurrent viral, fungal, and protozoal infections? They can usually handle pyogenic infections.
- T cell dysfunction
 - Clinical example: **DiGeorge syndrome**
- Also: **Wiskott-Aldrich** (microthrombocytopenia)
- Sort of T cells: **CMC, APS-1**
- TLR
 - Not T cells, but: **UNC93B** or **TLR3**



Slide 37. T-cell-dysfunction phenotype: viral/fungal/protozoal susceptibility is the pattern. The deck separates true T-cell defects from mimic mechanisms such as UNC93B/TLR3.

Hyper IgM Phenotype



- **Looks like antibody deficiency phenotype** (mostly the same kinds of infections, but some fungi)
- **Normal or elevated IgM** level
- Other classes are **very low** (approaching zero)
- Six known gene variants permitting a specific diagnosis; others await discovery
 - HIGM1: **CD40L** deficiency - high yield
- Not IgM, but: **NEMO** (a class switching defect)



Slide 38. Hyper-IgM phenotype: IgM normal/elevated while other immunoglobulin classes are very low. CD40L deficiency is highlighted as the high-yield specific cause.

Antibody Deficiency Phenotype



- Variable patterns of infection (bacterial mostly)
- Can do OK with the infections that need T cells, such as some viruses, fungi, and protozoa
- Antibody is an important opsonin
 - Clinical features of Ab deficiency can be similar to those of phagocyte or some types of complement deficiency (recurrent boils and abscesses)
- **XLA1, selective IgA deficiency, 'CVID' (TACI and BAFF-R, maybe APDS), transient hypogammaglobulinemia of infancy**
- Not antibody, but: **IRAK4** or **MyD88**; certain complement deficiencies



Slide 39. Antibody-deficiency phenotype: predominantly bacterial disease, but the deck warns that phagocyte/complement defects can mimic it because antibody is an important opsonin.

Slides 34-36: SCID Immunophenotypes

SCID Immunophenotypes



- T+ B+ NK+
 - p56Lck
- T- B+ NK- (B cells don't need these)
 - IL2RG (common gamma chain, γ c)
 - JAK3
- T- B+ NK+ (B cells and NK cells don't need these)
 - IL7RA
 - CD3 δ ; CD3 ϵ ; CD3 γ
- T- B- NK- (Metabolic defect)
 - ADA deficiency
- T- B- NK+ (NK cells don't have receptors with gene rearrangement)
 - RAG1 or RAG2; Artemis

The immunophenotype in Omenn syndrome is highly variable, even T+B-NK+



Slide 36. The faculty SCID table. This is one of the highest-yield slides in the lecture because it forces you to connect basic lymphocyte development to flow-cytometry phenotype.

Flow phenotype	Genes / defects listed on slide 36	Logic
T+ B+ NK+	p56Lck	Cells can be present despite profound signaling dysfunction
T- B+ NK-	IL2RG (common γ chain), JAK3	B cells do not require these pathways; T/NK development does
T- B+ NK+	IL7RA; CD3 δ , CD3 ϵ , CD3 γ	Selective disruption of T-cell development/signaling with preserved B/NK counts
T- B- NK-	ADA deficiency	Metabolic toxicity affects lymphocyte lineages broadly
T- B- NK+	RAG1/2, Artemis	V(D)J recombination needed by T and B receptors; NK cells do not rearrange antigen receptors

Ozzy checkpoint - click him once you can derive the T/B/NK pattern from the biology instead of memorizing isolated rows.

Omenn syndrome: the deck explicitly warns that the immunophenotype is highly variable, even T+B-NK+. Do not force every SCID syndrome into one memorized row.

Companion simplification: the Word guide teaches only three classic rows (IL2RG, ADA, RAG). For this course, use the broader slide-36 table above because it is the current faculty version.

Named IEIs - What Is Actually Supported Here

The current deck expects you to recognize many named IEIs, but it does *not* give a full dedicated disease slide for every name. The table below separates deck-emphasized facts from classic details supplied by the companion guide.

Common Variable Immunodeficiency



- **Common variable immunodeficiency (CVID)** is a very heterogeneous group of largely unclassified disorders presenting with the antibody deficiency phenotype
- 'CVID' is only a **working preliminary diagnosis** when:
 - 2 or more immunoglobulin classes are low, **and**
 - **There is no alternative explanation**
- Many specific causes of 'CVID' have been found in the molecular medicine era
 - Defects in **BAFF**, **TACI**, and **APRIL** are the highest yield.



Slide 40. CVID is explicitly taught as a heterogeneous antibody-deficiency phenotype and a working preliminary diagnosis: ≥ 2 low immunoglobulin classes with no alternative explanation.

IEI	Current-deck emphasis	Companion-only classic detail when useful
“CVID”	Heterogeneous phenotype; ≥ 2 low Ig classes + no alternative explanation; BAFF/TACI/APRIL high-yield; molecular diagnosis encouraged; gamma globulin replacement; APDS may have targeted leniolisib.	Often later presentation with recurrent sinopulmonary infection and poor vaccine response.
XLA	Antibody-deficiency phenotype; very low immunoglobulins; individual BTK test is a model of targeted genotyping.	BTK defect arrests B-cell development; absent mature B cells; boys often become symptomatic after maternal IgG wanes.
Selective IgA deficiency	Listed within antibody-deficiency phenotype.	Often asymptomatic; recurrent sinopulmonary/GI disease can occur; transfusion reaction risk is classic companion material.
DiGeorge	T-cell-dysfunction phenotype; abnormal TREC can detect congenital lymphopenia.	22q11.2 / thymic-parathyroid developmental syndrome details are companion enrichment.
Wiskott-Aldrich	Microthrombocytopenia on CBC; appears under T-cell-dysfunction phenotype.	Eczema + thrombocytopenia + recurrent infection triad is companion-supported.
CGD	DHR failure; phagocyte phenotype; hypergammaglobulinemia may occur; Aspergillus/Nocardia/Serratia/Staph clues.	NADPH oxidase / catalase-positive framing is companion-supported.
LAD	Phagocyte phenotype; delayed cord separation can occur but the deck	No-pus infections + neutrophilia + adhesion

	explicitly calls the LAD-I association an “urban legend” if overused diagnostically.	defect are classic companion details.
Complement deficiency	Terminal complement deficiency → recurrent Neisseria; other complement phenotypes include bacterial infection, lupus-like disease, PNH, angioedema.	Specific component-disease associations beyond this are secondary enrichment.
Hereditary / acquired angioedema	C1 inhibitor quantity/function in workup; C1q is listed for acquired angioedema.	Detailed bradykinin pathway treatment is not supplied by this lecture.
Hyper-IgM	Normal/high IgM with other classes nearly absent; CD40L deficiency is high-yield; NEMO can mimic a class-switch defect.	Word guide supplies more classic mechanism detail if needed.
Transient hypogammaglobulinemia of infancy	Listed within antibody-deficiency phenotype.	No dedicated current-deck disease slide; avoid adding unsupported detail to faculty content.
Chédiak-Higashi	CBC/smear clue and phagocyte phenotype list.	No dedicated current-deck disease-mechanism slide here.

Complement Deficiency Phenotypes



- Terminal complement deficiency (think George D. Heist): recurrent *Neisseria* infections
- Other presentations
 - Bacterial infections (similar to antibody deficiency phenotype)
 - Rheumatologic (immune complex; "lupus-like") diseases
 - Paroxysmal nocturnal hemoglobinuria
 - Hereditary and acquired angioedema



Slide 44. Complement-deficiency phenotype. Terminal complement deficiency is linked to recurrent *Neisseria*; other presentations broaden beyond infection alone.

Phagocyte Deficiency Phenotype



- Boils and abscesses (deep-seated; internal organs)
- Pneumonia and osteomyelitis
- Certain organisms are mostly dealt with by neutrophils
 - Infections with *Aspergillus*, *Nocardia*, *Serratia*, and *Staphylococcus* suggest neutrophil dysfunction
- Hypergammaglobulinemia, leukocytosis
- Delayed separation of the umbilical cord found in CGD, LAD-I, LAD-III, some others – but infection or keeping the site too clean are more common
- CGD, **LAD-I - LAD-II - LAD-III**, hyper IgE type 1, Chédiak-Higashi
- (LAD-I urban legend: delayed umbilical cord separation **denotes** LAD-I)



Slide 45. Phagocyte-deficiency phenotype. The organism pattern and deep abscesses matter more than reflexively anchoring on one single disease clue.

High-Yield Traps & Vignette Logic

Vignette / result	Best first thought from this lecture
Infant + failure to thrive + thrush + very low ALC	Combined/SCID phenotype → flow cytometry; check newborn TREC result
Normal/high IgM + near-zero IgG/ IgA/IgE	Hyper-IgM phenotype
Mostly recurrent bacterial sinopulmonary disease	Antibody deficiency phenotype , but remember phagocyte/complement mimics
Deep abscesses + Aspergillus/ Nocardia/Serratia/Staph	Phagocyte dysfunction → DHR is high-yield
Recurrent Neisseria	Terminal complement deficiency
Abnormal TREC	Congenital/secondary lymphopenia evaluation , not “SCID until proven otherwise” as a final diagnosis
“CVID” label	Phenotype first ; seek a specific molecular diagnosis when possible
Delayed umbilical separation	Can occur in LAD/CGD and others; faculty warns not to anchor on LAD-I alone

Summary

When you suspect IEI:

- Get a detailed history of the types of infections
- Think in terms of disease phenotypes
- Order a screening evaluation
- Narrow it down from there; **send an informed consult immunology early on**
- Genotyping is available for nearly all diseases
- Whole exome/genome sequencing will find new diseases
- Most patients will be “normal,” but some new IEIs await discovery (maybe by you)



Slide 50. The lecture's closing algorithm: detailed infection history → phenotype → screening evaluation → narrow the differential → early informed immunology consult → genotyping / exome-genome when appropriate.

FINAL ALGORITHM

Pattern of infections tells you which arm of host defense to test.

CBC + GAME + HIV + complement + TREC history → targeted flow/function → genetics.

Source hierarchy: Dolen 54-slide current deck is primary. The Word guide is secondary enrichment. Companion mnemonics and classic disease details are not silently treated as faculty-slide content.